Pablo Silva

# Analysis I - Chapter 2

## 2.1 - The Peano Axioms

## 2.2 - Addition

1. Let us induct on $c$ while $a$ and $b$ remain fixed. For the base case $c = 0$:
$(a+b)+0 = a+(b+0)$
By L 2.2.2 and by L 2.2.3:
$\boxed{(a+b) = (a+b)}$

Now we induct:
$(a+b)+c{+}{+} = a+(b+c{+}{+})$
By L 2.2.3 and D 2.2.4:
$((a+b)+c){+}{+} = a+(b+c){+}{+}$
$\Rightarrow ((a+b)+c){+}{+} = (a+(b+c)){+}{+}$ A2.4
$\Rightarrow \boxed{(a+b)+c = a+(b+c)}$
This closes the induction.

2. With base case $a=1$, the 1st positive number, $b=0$; $b{+}{+} = 1 = a$. Therefore we induct on $a$:
$a{+}{+}$ must therefore have $(b{+}{+}){+}{+} = a{+}{+}$. By A2.4 $\boxed{b{+}{+} = a}$. This closes the induction.

3. a. Base case $a=0$:
$0 \geq 0$; by D 2.2.11:
$0 = 0+c$; by D 2.2.1 and P 2.2.4:
$\Rightarrow 0 = c \Rightarrow c = 0$.
Now we induct:
$a{+}{+} \geq a{+}{+}$ by D 2.2.11:
$a{+}{+} = a{+}{+} + c$. Since $c=0$: $a{+}{+} = a{+}{+}$ and by D 2.2.11 and A2.4:
$\boxed{a \geq a}$

b. We have:
$a = b + x_1$ and $b = c + x_2$ for natural numbers $x_1, x_2$.
Therefore:
$a = (c + x_2) + x_1$. By P. 2.2.5:
$a = c + (x_1 + x_2)$. By P.2.2.8 and C.2.2.9, $(x_1+x_2)$ must also be some natural number. Therefore $x_3 = x_1 + x_2$ is a natural number:
$a = c + x_3 \Rightarrow \boxed{a \geq c}$

c. We have:
$a = b + x_1$ and $b = a + x_2$ where $x_1$ and $x_2$ are natural numbers. We induct on $a$.
Let $a = 0$:
$0 = b + x_1$ and $b = 0 + x_2$. By C.2.2.9, D.2.2.1, and P.2.2.4:
$b = x_1 = 0$. $b = x_2 = 0$.
Therefore $a = b$. Now for $a{+}{+}$:
$a{+}{+} = b + x_1{+}{+}$ By L.2.2.3
$a{+}{+} = (b+x_1){+}{+}$ and by A2.4:
$a = b + x_1$ with $b = a{+}{+} + x_2$ and by D.2.2.1 $b = (a+x_2){+}{+}$ which again has $b = a + x_2{+}{+}$.
This implies that $b$ was also incremented: $b{+}{+} = a + x_2{+}{+}$.

d. We have $a = b + x_1$ and $(a+c) = (b+c) + x_2$. We therefore have by L.2.2.3 and L.2.2.2:
$a + c = (b+c) + x_2$. By P.2.2.5:
$a + c = c + (b + x_2)$. By P.2.2.6 we have $a = b + x_2$. Thus,
$\boxed{a \geq b}$ iff $a + c \geq b + c$

e. If $a<b$, this implies $a \neq b$ by D.2.2.11 though by L.2.2.10 we could have some number $a{+}{+} = b$ where $a \neq b$. Therefore
$\boxed{a{+}{+} \leq b}$.

f. By D.2.2.11 we have $a<b \Rightarrow b>a$. Therefore we have $b = a+d$. The only way to get $b = a$ is if $d = 0$ by L.2.2.2, which is absurd. Therefore $d$ must not be zero — by D.2.2.7 $d$ must be a positive number.

4. I - $0 \leq b$ holds as by D.2.2.11: $b = 0 + d$ and by D.2.2.1 $b = d$. This is true for all $b$.

II. By D.2.2.12 & D.2.2.1:
$a = b + d$ for a pos. num. $d$.
Therefore $a{+}{+} = (b+d){+}{+}$ and by D.2.2.1 $a{+}{+} = b + d{+}{+}$. Since $n{+}{+} = n+1$ and D.2.2.7 & D.2.2.8, we have $a{+}{+} = b + e$ (pos. num.)

III - By D.2.2.11, $a \geq b$ and $a \leq b$. We have $a = b + d$ whereby $d = 0$. Therefore
$a{+}{+} = (b+d){+}{+} \Rightarrow a{+}{+} = b + d{+}{+}$.
By D.2.2.8 we have $d$ being a pos. num. Therefore D.2.2.12 applies:
$\boxed{a{+}{+} > b}$

5. Let $Q(n)$ be the property whereby $P(m)$ is true for all $m_0 \leq m < n$. By setup, $Q(0)$ will be true vacuously. As such, $Q(n+1)$ must therefore have $m_0 \leq m < n{+}{+}$, which by P.2.2.13 preserves the inequality. Therefore $P(m)$ now holds for all $m_0 \leq m < n{+}{+}$. Thus, we treat $m$ as $n'$ and $n{+}{+}$ as $m$ wherefrom $P(m)$ is defined to be true.

6. If $P(n)$ is true, we therefore have $P(m) = P(n)$ to also be true. Then, for all $m<n$, there must be by backwards induction one $m$ such that $m{+}{+} = n$. Therefore $P(m{+}{+})$ is also true and henceforth all $P(m)$ are true by definition. Note that this proof is vacuous for base case $n=0$.

7. We start at the base case $m=n$ from $m \geq n$. Since $P(m)$ is true, it therefore follows that $P(n)$ is true. Then, by induction, $P(m+1)$ is also defined to be true and as such all $m \geq n$ are true for $P(m)$.

## 2.3 Multiplication

1. We first prove $b \times 0 = 0$. With base case $b=0$:
$0 \times 0 = 0$ by D.2.3.1. Then inductively we have $(b{+}{+}) \times 0 = (b \times 0) + 0$ which we defined to be
$\Rightarrow (0) + 0 = \boxed{0}$. We now define $b \times a{+}{+} = (b \times a) + b$.
Inducting on $b$:
$0 \times a{+}{+} = 0$. Therefore
$b{+}{+} \times a{+}{+} = (b \times a{+}{+}) + a{+}{+}$
and thus $((b \times a) + b) + a{+}{+}$
which equals $((b \times a) + b + a){+}{+}$
and thus $((b \times a) + a + b){+}{+}$
Then, $0 \times b = 0 = b \times 0$ and
$a{+}{+} \times b = (a \times b) + b = b \times a{+}{+} = (b \times a) + b$
We therefore have $(a \times b) + b = (b \times a) + b$
and from P.2.2.6:
$$\boxed{(a \times b) = (b \times a)}$$

2. Suppose both $n = m = 0$:
$0 \times 0 = 0$. We define $n \times 0 = 0$. By induction,
$(n{+}{+}) \times 0 = (n \times 0) + 0$
$= 0 + 0 = 0$.
Thus $n \times 0 = 0$ and $0 \times m = 0$. If neither $n \neq 0$ and $m \neq 0$, we have $n{+}{+} \times m = (n \times m) + m$ which must be positive as $m$ is positive by P.2.2.8. Similarly if $n \times m{+}{+}$
$\Rightarrow m{+}{+} \times n = (m \times n) + n$.
Since $n$ is positive, by P.2.2.8 we have $(m \times n) + n$ is positive.

3. Let us maintain $a$ and $b$ fixed and induct on $c$.
$(a \times b) \times 0 = 0$ by L.2.3.?
Similarly, $a \times (b \times 0) = 0$.
Then, we induct on $c$:
$(a \times b) \times (c{+}{+}) = (a \times b) \times (c+1)$
$\Rightarrow ab(c+1)$
$\Rightarrow abc + ab$
$\Rightarrow ab[1 + cc]$
Then, $a \times (b \times (c{+}{+}))$
$\Rightarrow a \times (b \times (c+1))$
$\Rightarrow a \times (bc + b) = \boxed{abc + ab}$
Therefore $(a \times b) \times c = a \times (b \times c)$

4. $(a+b)^2 = a^2 + 2ab + b^2$
We thus defined exponentiation:
$(a+b)^2 = (a+b) \times (a+b)$
By D.2.3.4 we have:
$(a+b)a + (a+b)b$
$\Rightarrow a^2 + ba + ab + b^2$. By L.2.3.2, $ba = ab$ and thus
$(a+b)^2 = a^2 + 2ab + b^2$

5. $n = mq + r$. Let us fix $q$ and induct on $n$:
Setting $n = 0$:
$0 = mq + r$. By C.2.2.9 $mq = r = 0$.
$n{+}{+} = (mq + r){+}{+}$ By A.2.4 we therefore have $n = mq + r$. This closes the induction.